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Impact of Arsenic trioxide on the hematological parameters in Rabbit model

Manisha Kumari[a*] and Anil Kumar[b]

[a]Department of Zoology, Nirmala College, Ranchi University, Ranchi-834002, Jharkhand

[b]MDLM Hospital and Research Centre, Ranchi- 835217, Jharkhand

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Abstract : Arsenic is one of the most dangerous occupational and environmental toxins. Both natural and anthropogenic sources are responsible for the distribution of many toxicants, mainly heavy metals throughout the environment. Arsenic trioxide is a trivalent inorganic compound of arsenic. The animals were divided into three groups. Group I : treated as control. Group II: treated with 0.2mg/kg of As_2O_3 for 15 days and Group III : treated with 0.6mg/kg of of As_2O_3 for 7 days. In the arsenic treated groups there was a significant decrease in Hb%, PCV, MCV, MCH, and MCHC content.

Keywords : Arsenic, Hemoglobin, PCV, MCV

INTRODUCTION

Arsenic toxicity is a global health problem affecting many millions of people. Contamination is caused by arsenic from natural geological sources leaching into aquifers, contaminating drinking water and also from mining and other industrial processes. Absorption occurs predominantly from ingestion from the small intestine, though minimal absorption occurs from skin contact and inhalation.

Arsenic exerts its toxicity by inactivating up to 200 enzymes, especially those involved in cellular energy pathways and DNA synthesis and repair. Acute arsenic poisoning is associated initially with nausea, vomiting, abdominal pain, and severe diarrhea. Hematological abnormalities have been reported in acute and chronic arsenic poisoning[1]. A characteristic pattern of anemia, leucopenia and thrombocytopenia was found in 55 individuals exposed to arsenic in drinking water in Niigata Prefecture in Japan for Approximately 5 years, half of the subjects having arsenical skin lesion[2].In one study in West Bengal, anemia was reported in all the 13 people exposed to arsenic contaminated groundwater(0.2-2mg/l)[3].Arsenicosis is defined as a chronic health condition arising from prolonged ingestion of arsenic above the safe dose for at least six months usually manifested by characteristic keratosis, occurring alone or in combination with or without the involvement of internal organs[4].

*Corresponding author :
Phone : 919471518815
E-mail : ranchimanishakumari@gmail.com

MATERIAL AND METHODS

Mature and healthy Rabbits belonging to Order – Lagomorpha Family – Leporidae were used to assess the effect of Arsenic toxicity. Rabbits weighing from 1.50 – 2.25kgs were obtained from the Veterinary College (Birsa Agricultural University), Kanke, Ranchi. They were kept in cages and supplied with dechlorinated tap water for acclimatization at 30.2 - 34.5^{0}C for fourteen days, during which they were fed with green leafy vegetables and grains *ad libitum*. The natural photo period was maintained during the period. Mortality during the experiment of acclimatization was less than 2%. Group I was maintained as control. Group II was exposed to 0.2mg/kg As_2O_3 for the period of 15 days. Group II was exposed to 0.6mg/kg As_2O_3 for the period of 7 days intraperitoneally. Blood samples for baseline data and of other treatments were collected from the ear vein of each rabbit with syringes and needles. Samples were stored in vials without EDTA and with EDTA (10%) in deep freeze and were analyzed in the laboratory for various liver parameters by standard

techniques.

RESULTS

The data was statistically analyzed both by T-test and ANOVA through which it was observed that the complete picture (CP) of liver enzymes were significantly disturbed in arsenic feed groups. In the hematological tests (Table1) the hemoglobin percentage (Hb%) of group II (treated with 2mg/kg of As_2O_3) decreased to a mean value of 7.23 g/dl ±0.52 while that of group III (treated with 6mg/kg of As_2O_3) rabbits had become come down to 6.98 g/d± 0.67 which is much lesser than the group I value of 12.5 g/dl. Since the P-value of the F-test is less than 0.05, there is a statistically significant difference between the means of the 3 variables at the 95.0% confidence level. The PCV value of group II rabbits was decreased to a mean value of 25g/dl±2.23 while that of group III rabbits values even comes more significantly lesser about 18 g/dl±1.89 than the group I value of 40.0 g/dl, The F-ratio, which in this case equals 219.375, is a ratio of the between-group estimate to the within-group estimate. Since the P-value of the F-test is less than 0.05, there is a statistically significant difference between the means of the 3 variables at the 95.0% confidence level. . The MCV value of group II comes 62.92±6.83 and group III comes 69.92±8.23.Since the P-value of the F-test is greater than or equal to 0.05, there is not a statistically significant difference between the means of the 3 variables at the 95.0% confidence level.

The MCH value of group II was recorded 19.38 pg±3.25 but in group III the values became significantly lesser than group I value which is 16.24 pg±4.28. Since the P-value of the F-test is greater than or equal to 0.05, there is not a statistically significant difference between the means of the 3 variables at the 95.0% confidence level.

The MCHC value of group II animals went down to about 27.92 g/dl±6.32 still group III value were much more lesser i.e. 25.82 g/dl±8.26 than group I value of 34.05 g/dl. Since the P-value of the F-test is greater than or equal to 0.05, there is not a statistically significant difference between the means of the 3 variables at the 95.0% confidence level

TABLE : Hematological Data (All values are Mean + SEM)

	Group I	Group II	Group III
Hb%(g/dl)	12.5± 1.7(10-15)	7.23± 0.52	6.98± 0.67
PCV%(g/dl)	40.0± 1.5(30-50)	25± 2.23	18±1.89
MCV(fl)	56.0± 1.9(58-79)	62.92± 6.83	69.92±8.23
MCH(pg)	21.0± 1.6(19-29)	19.38±3.25	16.24± 4.28
MCHC(g/dl)	34.05± 0.6(31-37)	27.92± 6.32	25.82± 8.26

2. Graphs

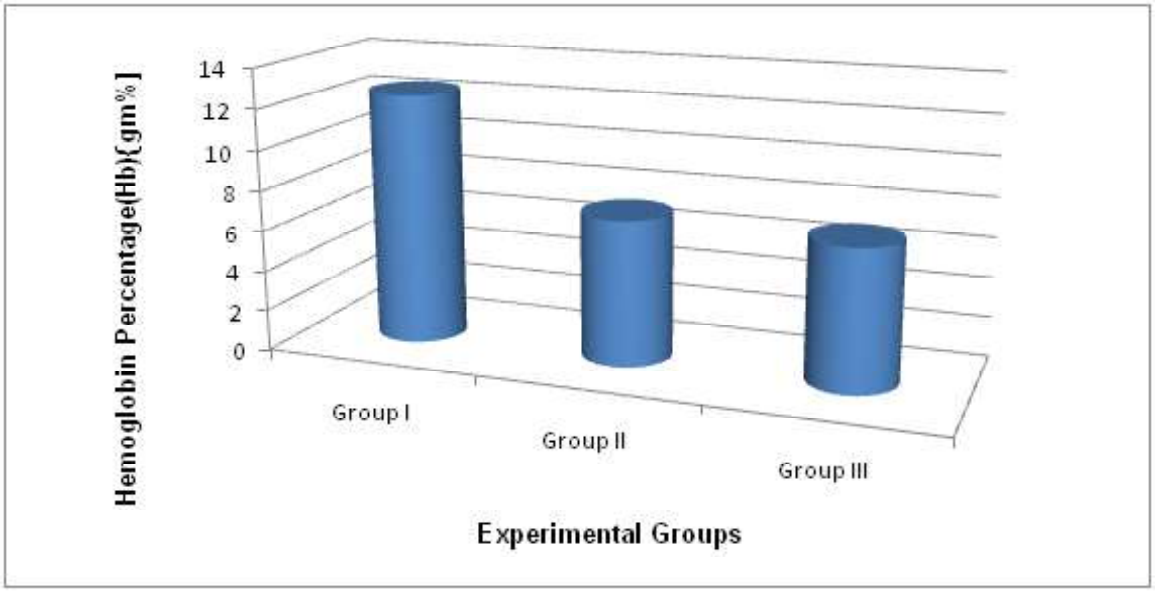


Fig 2.1 Hemoglobin percentage of Rabbit of three different groups. Group I, Group II and Group III. Bars are Mean±(n=5) from the respective control group

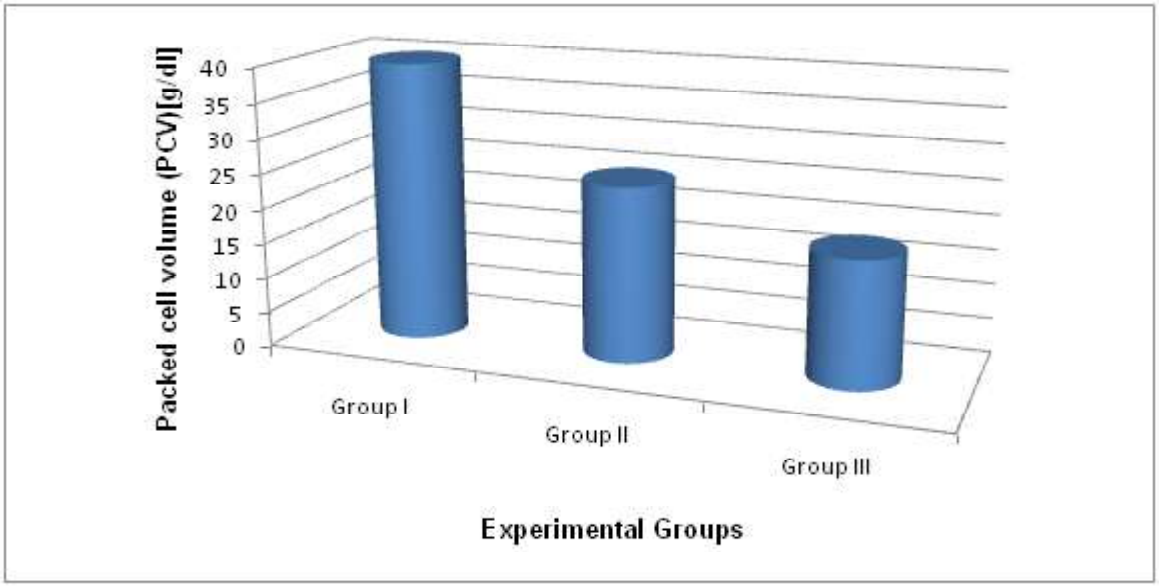


Fig 2.2 Packed cell volume of Rabbit of three different groups. Group I, Group II and Group III. Bars are Mean±(n=5) from the respective control group

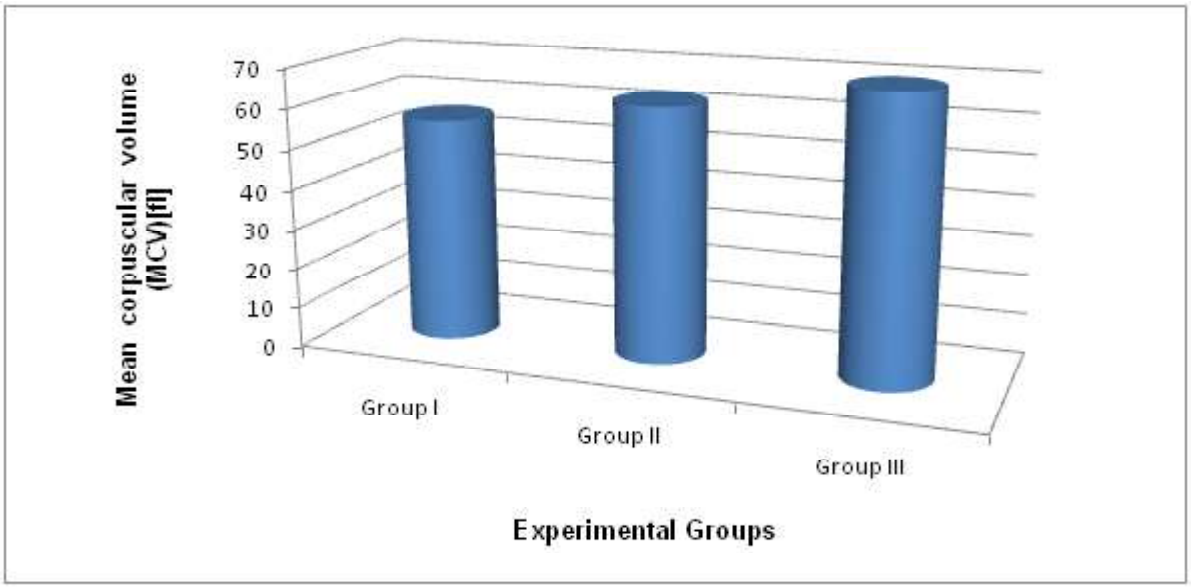


Fig 2.3 Mean cell volume of Rabbit of three different groups. Group I, Group II and Group III. Bars are Mean±(n=5) from the respective control group

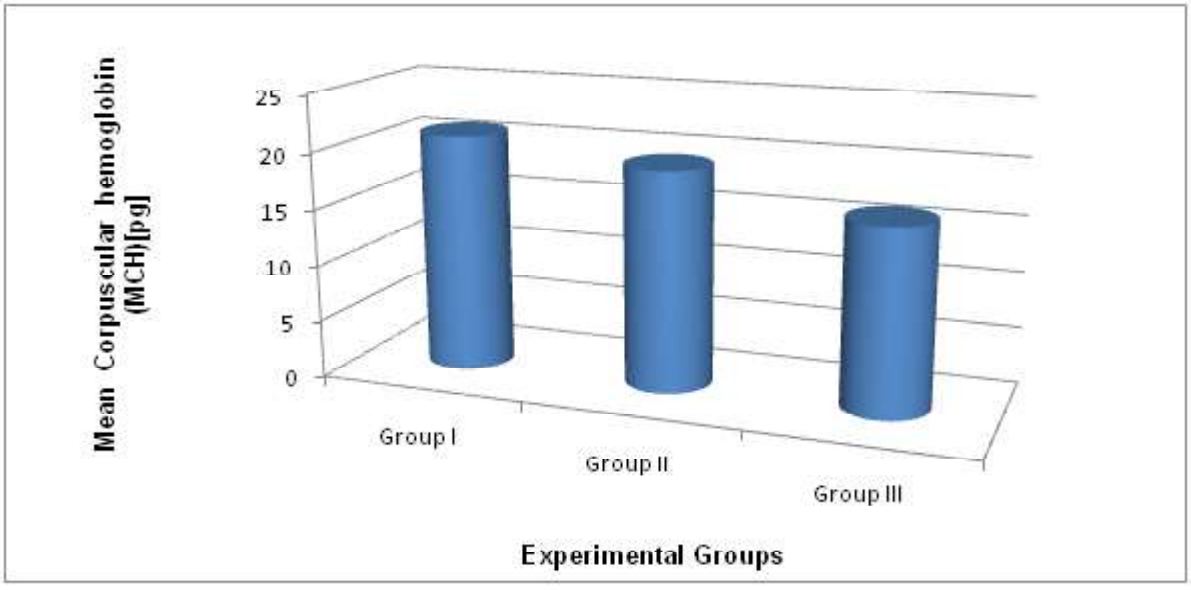


Fig2.4 Mean corpuscular hemoglobin of Rabbit of three different groups. Group I, Group II and Group III. Bars are Mean±(n=5) from the respective control group

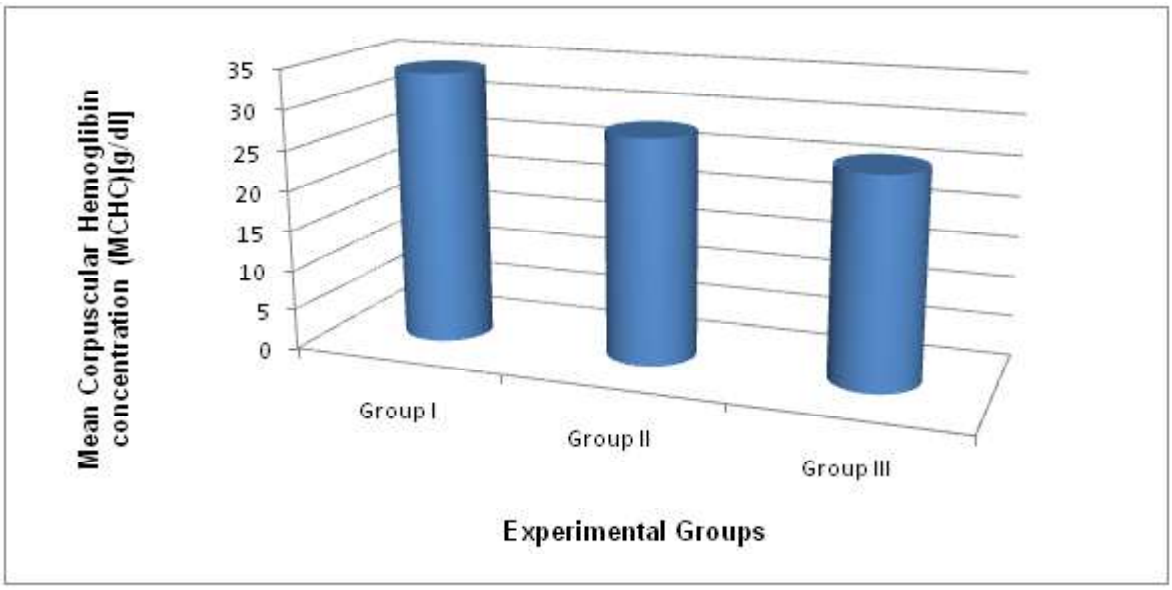


Fig2.5 Mean cell hemoglobin concentration of Rabbit of three different groups. Group I, Group II and Group III. Bars are Mean±(n=5) from the respective control group

DISCUSSION

Previous studies[5]revealed that WBC level decreased when mice were given higher dose of arsenic. This might be due to apoptotic effect of arsenic on plasma cells. The hemoglobin content gets decreased in the arsenic treated rabbits of the present study as compared to control group. In previous works[6,7] it was found that the RBC and hemoglobin level is decreased with increased concentration of arsenic. This could be due to binding ability of arsenic to hemoglobin that leads to inhibition of haem synthesis pathway[8].

Arsenic exposure has also been known to influence the activity of several enzymes of haem biosynthesis[8]. It has been reported that arsenic exposure produces a decrease in ferrochelatase and decrease in COPRO-OX and increase in hepatic5 aminolevulinic acid synthetaseactivity[9,10].

The haem synthesis pathway plays an important role in all nucleated cells to provide chlorophyll and related structure[11,12]. In mammalian and avian tissues the principle product of this pathway is haemferroprotoporphyrin IX, an essential component of various biological functions including oxygen transport systems and metabolic processes. This pathway is known to be highly susceptible to alterations induced by environmental pollutants offering the opportunity to use these changes as indicators of damage caused by many metals or metalloids. About 85% of haem are synthesized in bone marrow where it is required for hemoglobin formation. Remaining 15% is synthesized in liver and other organs where it is required for hemoprotein synthesis. Since most metals have strong affinity for nucleophilic ligands, each step of the haem biosynthesis pathway is potentially susceptible to direct inhibition as a result of metal mercaptide bond formation with the functional sulfhydryl groups. A number of studies have demonstrated that porphyrins and other constituents of the haem synthesis pathway might serve as sensitive and specific biomarkers of toxic metal exposure in humans. It was observed during the study that RBCs, WBCs and hemoglobin level were disturbed then automatically other parameters like Mean cell volume, Mean cell hemoglobin, and Mean cell hemoglobin concentration were also be disturbed because these are dependent on the former parameters. In a previous study[13] on mice the reduction in WBCs have thought to have resulted in the reduction in HCT. The very significant reduction in MCV in this study caused by 4.0-4.8mgL^{-1} arsenic exposure implies that the RBCs had become microcytic. Microcytosis will result in the RBCs not being able to carry enough oxygen for efficient tissue respiration[14].This could result in ATP deficit (energy reduction),general body weakness and death[13].An earlier study has shown a moderate hemolytic anemia linked

to arsenic exposure in mice[15,16].

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